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DETAIL SPECIFICATION

BLOWERS, MINIATURE, FOR COOLING ELECTRONIC EQUIPMENT GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers alternating current (ac) fractional horsepower miniature blowers having specific environmental characteristics required for cooling communication, special military equipment, and other electronic equipment. The blowers described herein are capable of withstanding extreme service and environmental conditions such as shock, high temperature, low temperature, vibration, humidity and fungus.

1.2 Classification.

1.2.1 Part or identifying number (PIN). The part or identifying number (PIN) consists of the letter "M", the basic number of the specification sheet, and an assigned dash number as shown in the following example (see 3.1).

M23071	/1	-001
Military designator	Specification sheet number	Dash number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28	Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-16878	Wire, Electrical, Insulated, General Specification for

(See ASSIST database for list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	Military Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Electronic and Electrical Component Parts
MIL-STD-202-101	Method 101, Salt Atmosphere (Corrosion)
MIL-STD-202-201	Method 201, Vibration
MIL-STD-202-211	Method 211, Terminal Strength
MIL-STD-202-213	Method 213, Shock (Specified Pulse)
MIL-STD-202-302	Method,302, Insulation Resistance
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-2073-1	Standard Practice for Military Packaging
MIL-STD-13231	Marking of Electronic Items

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454	General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL

AMCA 210	Laboratory Methods of Testing Fans for Certified Aerodynamic Performance Rating
AMCA 300	Reverberant Room Methods for Sound Testing of Fans

(Copies of these documents are available from https://www.amca.org Air Movement and Control Association International, Inc.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASME PTC 19.5 Flow Measurement, Performance Test Codes

(Copies of this document are available from https://asme.org American Society of Mechanical Engineers, Three Park Avenue, M/S 10E, New York, NY 10016-5990.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 Procedures, Sampling and Tables for Inspection by Attributes.

(Copies of this document are available from https://asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE 119 General Principles of Temperature Measurement as Applied to Electrical Apparatus, IEEE Recommended Practice for

(Copies of these documents are available online at https://ieeexplore.ieee.org or the Institute of Electrical and Electronic Engineers, 445 Hoes Lane, Piscataway, NJ 08855-1331.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA MW1000 Magnetic Wire

(Copies of these documents are available online at https://www.global.ihs.com.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE AS50151 Connectors, Electrical, Circular Threaded, AN Type, General Specification for

(Copies of these documents are available from www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specifications sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.3.1 Flammable and explosive material. When specified in the contract or purchase order (see 6.2), blowers shall contain the smallest possible amount of flammable and explosive material consistent with good design practice.

3.3.2 Corrosive material. Corrosive material used in any of the processes in the manufacture of blowers shall be neutralized to minimize corrosion in the completed blower.

3.3.3 Fungus and moisture resistant materials. Fungus inert materials in accordance with Guideline 4 of MIL-HDBK-454 and moisture resistant materials should be used.

3.3.4 Wire.

3.3.4.1 Magnet. Magnet wire shall be in accordance with NEMA-MW1000.

3.3.4.2 Leads. Wire leads shall be in accordance with MIL-DTL-16878. Soldering should meet Guideline 5 of MIL-HDBK-454.

3.4 Design and construction.

3.4.1 Motor frame and housing. Unless otherwise specified (see 3.1), the motor frame and housing shall be constructed of aluminum alloy and shall be finished with sufficient protective coating to pass the environmental tests (see 4.7.16 through 4.7.20 and 4.6.2).

3.4.1.1 Blower housing. The blower housing shall be smoothly finished in order to minimize audible noise caused by the airflow and shall be securely attached to the motor. The blade housing, when used as the normal means of support for the item, shall have strength to withstand the vibration and shock tests of MIL-STD-202, (see 4.7.17 and 4.7.18). Unless otherwise specified (see 3.1), the blade housing finish shall be black.

3.4.2 Threaded parts. All threaded parts shall be in accordance with FED-STD-H28 and shall be of such design and construction as to remain mechanically and electrically (where applicable) secure for the entire life of the blower and under conditions of the environmental requirements specified herein (see 4.7.16 through 4.7.20). Dissimilar metals shall not be used unless suitably protected against electrolytic corrosion.

3.4.3 Locking of screw-thread assemblies. All screw-thread assemblies shall be rendered resistant to loosening under vibration.

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3.4.4 Electrical Connections. Unless otherwise specified (see 6.2), electrical connections shall be made with the use of connectors conforming to MIL-DTL-26482, type MS-3112E-8-4P. The mating connector is MS-3116E-8-4S. For motors having wire size AWG 12, or larger, the connector shall conform to SAE-AS50151, type MS-3102R-18-11P.

3.4.5 Weight. The maximum weight of the blower shall not exceed the value specified (see 3.1).

3.4.6 Mounting. Unless otherwise specified (see 3.1), blowers shall be capable of being mounted in any orientation. Blowers having identical mountings and silhouette dimensions will have the same specification sheet number but different dash numbers (where applicable).

3.5 Performance requirements. Unless otherwise specified (see 3.1), blower performance tests including maximum power, current and temperature rise shall be conducted at 0 inches water column (see 6.4) static pressure and at the rated voltage and frequency as specified (see 3.1). Prior to testing, all blowers shall be subjected to a warm-up period. The run-in period for continuous duty blower shall be a minimum of 1 hour. Warm-up period between each of the tests is not required provided the time lapse between the preceding and succeeding tests does not exceed three hours.

3.5.1 Wire lead stress. All connections shall be mechanically secure and electrically continuous prior to soldering and shall be tested as specified in 4.7.2.

3.5.2 Input current. When measured as specified in 4.7.3, the input current (running and lock rotor) shall not exceed the value specified (see 3.1).

3.5.3 Input power. When measured as specified in 4.7.4, the input power shall not exceed the value specified (see 3.1).

3.5.4 Speed. Blower impeller speed shall be measured as specified in 4.7.5.

3.5.4.1 Speed (reference). The reference speed (or initial value) shall be recorded for future reference. The nominal speed indicated on the applicable specification sheet (see 3.1) shall approximate the reference speed.

3.5.4.2 Speed (subsequent). All subsequent speeds measured after the reference speed shall be not less than 95 percent of the reference speed of 3.5.4.1.

3.5.5 Burn-in. Burn-in shall be performed as specified in 4.7.6. At the end of the burn-in test, the speed test in accordance with 3.5.4 shall be performed to determine failure. A failure occurs when speed drops 5 percent below the initial speed test.

3.5.6 Generated vibration. When measured as specified in 4.7.7, the generated vibration of the blower shall not exceed the root-mean-square (rms) force pounds specified (see 3.1).

3.5.7 Line voltage and frequency. When tested as specified in 4.7.8, the change in blower speed shall be as specified (see 3.1).

3.5.8 Transient voltage test. The blowers shall be capable of withstanding voltage pulses of plus and minus 20 percent of steady state voltage as specified in 4.7.9. Pulse duration shall be 2 ± .05 seconds. After testing as specified in 4.7.9, blower performance shall be as specified in 3.5.4.2 and 3.5.2.

3.5.9 Dielectric withstanding voltage. When measured as specified in 4.7.10, there shall be no insulation breakdown or arcing.

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3.5.10 Insulation resistance. When tested as specified in 4.7.11, the insulation resistance shall be greater than 12 megohms.

3.5.11 Airflow performance. When tested as specified in 4.7.12, the blowers' airflow performance at two required static pressures shall be equal to or greater than that specified (see 3.1 and 6.4).

3.5.12 Dynamic balance (when applicable see 3.1). When tested as specified in 4.7.13, the blowers shall meet the limit specified.

3.5.13 Acoustic noise (when applicable see 3.1). When tested as specified in 4.7.14, the noise level in decibels (sound pressure level) at 3 feet shall not exceed the limit specified.

3.5.14 Electromagnetic compatibility. When specified, the blowers shall meet the limit requirements of MIL-STD-461 for surface ships tests CE101, CE102, CS116, RE101, RE102, RS103 (see 4.7.15 and 6.2).

3.5.15 Cold start. Unless otherwise specified (see 6.2), after 5 minutes of excitation at minimum ambient temperature, speed shall be at least 80% of reference speed in 3.5.4.1. After exposure as specified in 4.7.16, the blowers shall meet the requirements specified in 3.5.2 and 3.5.3 and speed as measured in 3.5.4.2.

3.5.16 Vibration. When tested as specified in 4.7.17, the blowers shall be visually and mechanically inspected and shall meet the requirements specified in 3.5.2, 3.5.4.2 and 3.5.6.

3.5.17 Shock. When tested as specified in 4.7.18, the blowers shall be visually and mechanically inspected and shall meet the requirements specified in 3.5.2, 3.5.4.2 and 3.5.6.

3.5.18 Humidity. When tested as specified in 4.7.19, the blowers shall be visually and mechanically inspected and shall meet the requirements of 3.5.2, 3.5.4.2, 3.5.9, and 3.5.10.

3.5.19 Salt spray. When tested as specified in 4.7.20, there shall be no evidence of significant corrosion on any part of the blower. The blowers shall be visually and mechanically inspected and shall meet the requirements of 3.5.2, 3.5.4. 2, 3.5.9, and 3.5.10.

3.5.20 Temperature rise. When tested as specified in 4.7.21, the winding temperature rise shall not exceed the value specified (see 3.1).

3.5.21 Extended life. Unless otherwise specified (see 3.1), the life of the blower tested shall be greater than 1000 hours when tested as specified in 4.7.22. The life of the blower shall be considered terminated when it fails to meet the speed requirements of 3.5.4. Input current, subsequent speed and generated vibration shall be measured as specified in 3.5.2, 3.5.4.2 and 3.5.6 after 1000 hours minimum.

3.6 Marking. Blowers shall be marked in accordance with MIL-STD-130 with military part number, manufacturer's name or symbol, capacitor value, terminal identification, wiring diagram, voltage, frequency, phase and current, air delivery (cubic feet per minute (CFM)) at zero static pressure (see 6.4), revolution per minute of motor, and maximum operating temperature. Markings shall be applied by acid or electric etching, permanent marking ink, or engraving applied directly to the housing or identification label. The markings shall be as permanent as the normal life expectancy of the blower on which it is applied and shall withstand the same environmental tests required of the blower (see 4.7.16 through 4.7.20). Metal decals conforming to MIL-STD-13231 may be used.

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3.7 Workmanship. The blowers shall be smoothly finished. Burrs and sharp edges shall be removed to minimize damage to personnel and to inhibit audible noise.

4. VERIFICATION

4.1

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3)
b. First article inspection (see 4.5).
c. Conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the blowers are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 Inspection conditions. Unless otherwise specified herein, all measurements and tests shall be made at an ambient temperature of 25°C ±5°C, atmospheric pressure of 30 ±2 inches of mercury, and a relative humidity no greater than 55 percent. The room ambient temperature change shall not exceed a rate of 2°C per hour. For environmental tests (see 4.7.16 through 4.7.20), temperature rate of change shall be approximately 3°C per minute.

4.5 First article inspection. When specified (see 3.1 and 6.2), first article inspection shall be performed.

4.5.1 Sample size. Two blowers shall be subjected to first article inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table I, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided as specified in table I for groups II and III.

4.5.3 Failures. One or more failures shall be cause for rejection.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all blowers of the same part number produced under essentially the same conditions and offered for inspection at one time.

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TABLE I. First Article Inspection.

Inspections	Requirement paragraph	Test method paragraph
Group I (two samples)		
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Wire lead stress	3.5.1	4.7.2
Input current	3.5.2	4.7.3
Input power	3.5.3	4.7.4
Speed	3.5.4	4.7.5
Burn-in	3.5.5	4.7.6
Speed	3.5.4	4.7.5
Dynamic balance	3.5.12	4.7.13
Line voltage and frequency	3.5.7	4.7.8
Transient voltage	3.5.8	4.7.9
Speed (subsequent)	3.5.4.2	4.7.5
Input current	3.5.2	4.7.3
Dielectric withstanding voltage	3.5.9	4.7.10
Insulation resistance	3.5.10	4.7.11
Group II (one sample)		
Airflow performance	3.5.11	4.7.12
Acoustic noise	3.5.13	4.7.14
Electromagnetic compatibility	3.5.14	4.7.15
Cold start	3.5.15	4.7.16
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Input power	3.5.3	4.7.4
Vibration	3.5.16	4.7.17
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Generated vibration	3.5.6	4.7.7
Shock	3.5.17	4.7.18
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Generated vibration	3.5.6	4.7.7
Humidity	3.5.18	4.7.19
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Dielectric withstanding voltage	3.5.9	4.7.10
Insulation resistance	3.5.10	4.7.11

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TABLE I. First Article Inspection. (continued)

Inspections	Requirement paragraph	Test method paragraph
Group II (continued)		
Salt spray	3.5.19	4.7.20
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Dielectric withstanding voltage	3.5.9	4.7.10
Insulation resistance	3.5.10	4.7.11
Group III (one sample)		
Temperature rise	3.5.20	4.7.21
Continuous duty motors	3.5.20	4.7.21.1
Extended life	3.5.21	4.7.22
Speed (reference and subsequent)	3.5.4	4.7.5
Input current	3.5.2	4.7.3
Generated vibration	3.5.6	4.7.7

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table II, in the order shown.

4.6.1.2.1 Sample plan. Group A inspections shall be performed on 100 percent of the production blowers. Major and minor defects shall be as defined in ASQ Z1.4.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1
Dielectric withstanding voltage	3.5.9	4.7.10
Input current	3.5.2	4.7.3
Speed (reference)	3.5.4.1	4.7.5
Dynamic Balance	3.5.12	4.7.13
Insulation resistance	3.5.10	4.7.11
Burn-in	3.5.5	4.7.6
Speed (subsequent)	3.5.4.2	4.7.5

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4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection. Prior to testing, all blowers shall be subjected to a warm-up period as specified in 3.5.

4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with table IV. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection.

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Airflow performance	3.5.11	4.7.12
Line voltage and frequency	3.5.7	4.7.8
Temperature rise	3.5.20	4.7.21
Wire lead stress	3.5.1	4.7.2
Cold start	3.5.15	4.7.16
Input current	3.5.2	4.7.3
Speed (subsequent)	3.5.4.2	4.7.5
Input power	3.5.3	4.7.4

4.6.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract, if the lot is accepted and the sample units are still within specified electrical tolerances.

TABLE IV. Sampling for group B inspections.

Number of blowers in lot	Number of blowers in sample
40 or less	2
41 to 110	3
111 to 300	4
301 to 500	6
501 to 800	10
801 to 3200	15
3201 and greater	25

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4.6.2 Inspection of packaging. Except when industrial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with group A and B quality conformance inspection requirements of MIL-STD-2073-1. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the requirements of MIL-STD-129. The inspection of industrial packaging shall be as specified in the contract (see 6.2)

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. The blowers shall be visually and mechanically inspected to ensure conformance with the specification (see 3.1, 3.3, 3.4, 3.6, 3.7).

4.7.2 Wire lead stress (see 3.5.1). Each wire shall be tested in accordance with MIL-STD-202-211, test condition C. Unless otherwise specified (see 3.1), weight of load shall be 5 pounds. Any indication of the lead pulling loose, wire strands breaking, or a permanent deformation of the insulation shall be cause for rejection.

4.7.3 Input current (see 3.5.2). The input current shall be measured by means of an ammeter.

4.7.4 Input power (see 3.5.3). The input power shall be measured by means of a wattmeter.

4.7.5 Speed (see 3.5.4). Impeller speed shall be measured using a stroboscope. The accuracy of the equipment used for measuring blower speed shall be with ±2 percent of the measured value.

4.7.6 Burn-in (see 3.5.5). The blowers shall be placed in an oven at the high-test temperature specified for 48 hours with nominal voltage and frequency specified (see 3.1).

4.7.7 Generated vibration (see 3.5.6).

4.7.7.1 Blower mounting. The blower shall be positioned with the shaft horizontal and elastically supported to allow vibration in all axes. The elastic members of the mounting shall be either tension or compression types and shall be directly attached to the blower motor. Cradles and other mounting devices shall not be used. The weight of the suspended blower, accelerometer, and accelerometer adapting block, if used, shall cause the elastic support to compress or stretch at least the amount given by the following formula:

$$d = \frac{880{,}000}{N^2}$$

where: d = minimum compression or stretch, inches

N = blower motor speed, revolution per minute (rpm)

4.7.7.2 Accelerometer mounting. The accelerometer shall be attached directly to the motor frame between the bearings (see applicable specification sheets) with its sensing axis perpendicular to the motor shaft. An intermediate adapting block may be used to match mating surfaces. Accelerometer and adapting block, if used, shall be attached by stud mounting, Eastman 910 cement, or other equally rigid adhesives. With blowers operating at speeds greater than 6000 rpm only stud mounting or Eastman 910 cement, or equal, shall be used. The combined weight of the accelerometer, mounting stud, and adapter block shall not exceed 1.60 ounces.

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4.7.7.3 Instrumentation. A calibrated accelerometer system output shall be fed directly into a rms voltmeter having an input impedance of at least 1.0 megohm (see 6.4). Conditioning of the accelerometer output by filtering or any other means is not permitted.

4.7.7.4 Root mean measure (rms) force calculation. The blowers shall be operating at least 15 minutes prior to testing. The mean steady state rms accelerometer in g's shall be recorded. The blower weight in pounds shall be determined and recorded. The weight in pounds of the accelerometer, mounting stud, and adapting block, if used, shall be determined, and recorded. The blower rms force output from the following formula shall be determined and recorded.

$$Frms = wa$$

where: w = combined weight of blower, mounting stud, and adapter block, pounds

a = measured acceleration, g-rms.

4.7.8 Line voltage and frequency (see 3.5.7). The blowers shall be operated at the extreme limits of the steady state band in combinations of voltage and frequency shown in table V. Speed measurements shall be taken at each limit.

TABLE V. Steady state voltage and frequency.

Condition	Lower limit (-10%)	Voltage rated	Upper limit (+10%)	Lower limit (-5%)		Frequency rated	Upper limit (+5%)
A (reference condition)		X				X	
B	X			X			
C			X				X
D			X	X			

4.7.9 Transient voltage (see 3.5.8). With the blower operating at conditions B and D (table V), five voltage pulses, at 10 second intervals, shall be superimposed on the steady state nominal voltage.

4.7.10 Dielectric withstanding voltage (see 3.5.9). A test potential of 980 +0, -30 volts rms, 60 hertz shall be applied from each winding to frame and between windings. The test potential shall be raised slowly (minimum time 3 seconds) to the specified value and maintained for one minute, then reduced slowly (minimum time 3 seconds) to zero. A failure shall be indicated by a leakage current in excess of one milliampere peak or the indication of internal arcing or corona discharge. The leakage current limit of one milliampere peak shall not include the current drawn by the equipment capacitance, or the capacitance of the windings under test. Subsequent dielectric tests will be at 880 +0, -30 volts rms.

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4.7.11 Insulation resistance (see 3.5.10). Unless otherwise specified (see 3.1), the blower shall be tested in accordance with MIL-STD-202-302, condition B. The test voltage shall be applied between ground (frame) and each winding which is not permanently grounded, with all other windings grounded. Windings which are normally connected together shall be considered as one winding for this test.

4.7.12 Airflow performance (see 3.5.11). Blowers intended for application at approximately sea level atmospheric pressure conditions shall be measured in terms of CFM with any apparatus applying the principle of nozzles generally described in ASME PTC 19.5, section 5 or AMCA 210. Other air volume rate monitoring instrumentation such as Latimer Flow elements, etc., may be used only upon written approval from the activity responsible for first article inspection.

4.7.13 Dynamic balance (see 3.5.12).

4.7.13.1 Mounting. The blowers shall be elastically mounted to allow vibration in all axes. A cradle may be used to support the blower in the elastic mounting providing its weight is not greater than 15 percent of the sum of the weight of the accelerometer, mounting block, if used , and blower. The elastic members may be in tension or compression but must deflect under the weight of the blower and its cradle, if used, an amount at least equal to the amount given in the following formula:

$$d = \frac{880{,}000}{N^2}$$

where: d = minimum stretch or compression, inches

N = blower speed, rpm

The deflection should not be so great that the elastic members is exceeded or that the limit of deflection is reached (that is, elastic members must not “bottom out”).

4.7.13.2 Instrumentation. A calibrated accelerometer system output shall be fed through a suitable filter which is tuned to read the vibration at the running speed of the blower. Since vibration frequencies below the running speed are not expected, a low-pass filter can be used. The output of the filter shall not attenuate or amplify the input signal from the accelerometer system, or, if it does, an appropriate correction factor must be determined for the frequency being measured and applied to the voltmeter reading. The output signal from the filter shall then be fed to a rms voltmeter with an input impedance of at least 1.0 megohm.

4.7.13.3 Measurements. The accelerometer readings are to be taken on the bearing housings in a plane perpendicular to the blower’s rotational axis or, if the bearing housings are not accessible, the vibration shall be measured at the extremities of the blower in a plane perpendicular to the blower’s rotational axis. The accelerometer may be mounted permanently to the blower or its cradle or it can be equipped with a pointed rod or screw and held with fingers pointed end against the blower. Convert the voltmeter reading using the accelerometer system sensitivity given as millivolts rms per G rms to G’s rms and, using the following formula, calculate the velocity reading in inches per second rms.

$$V_{rms} = \frac{3690 \text{ G rms}}{\text{Blower rpm}}$$

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4.7.14 Acoustic noise. Acoustic noise measurements are to be made in accordance with AMCA 300. The blower shall be resiliently supported as specified in 4.7.13.1 for measuring dynamic balance. The noise test is to be performed with the blower running at 0 inches water column static pressure.

4.7.15 Electromagnetic compatibility (see 3.5.14). Tests to determine conformance to the requirements of 3.5.14 shall be performed in accordance with MIL-STD-461.

4.7.16 Cold start (see 3.5.15). The non-energized blower shall be subjected to the minimum ambient temperature as specified (see 3.1) for a minimum of 16 hours. The blower, while at the low temperature, shall then be continuously energized at rated excitation for a period of 8 hours. Five minutes after initial excitation is applied, the blower speed shall be measured as described in 4.7.5. The blower shall then be deenergized and the above cycle repeated one more time.

4.7.17 Vibration (see 3.5.16). Unless otherwise specified (see 3.1), sample units shall be rigidly mounted on the vibration equipment utilizing the normal blower mounting surface and subjected to the vibration requirements of MIL-STD-202-201.

4.7.18 Shock (see 3.5.17). Unless otherwise specified (see 3.1), the non-energized sample units shall be securely mounted to the shock mounting fixture utilizing the normal blower mounting surface and subjected to the requirements of MIL-STD-202-213, figure 213-1, test condition A.

4.7.19 Humidity (see 3.5.18).

4.7.19.1 Nonoperating humidity cycling. The non-energized blowers shall be subjected to five 24-hour temperature variations consisting of approximately 16 hours at 60°C ±3°C and 8 hours at 30°C ±3°C. The relative humidity shall be maintained above a minimum of 95 percent during the steady state conditions. The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is 16 hours.

4.7.19.2 Operating humidity cycling. After remaining inoperative for not less than 12 hours or more than 24 hours at a temperature of 25°C ±3°C and 50 percent minimum relative humidity, the blower shall be operated continuously at rated voltage and frequency in an ambient temperature of 50°C ±3°C for a period of 8 hours. The relative humidity shall be increased to 95 percent minimum during the first 2 hours and maintained at 95 ±5 percent for 4 hours, then reduced to the conditions of 4.4 during the final 2 hours.

4.7.20 Salt spray. The blowers shall be tested in accordance with MIL-STD-202-101, condition B and with 5 percent salt solution. The blower shaft shall be horizontal when the blower is placed in the test chamber.

4.7.21 Temperature rise (see 3.5.20).

4.7.21.1 Continuous duty blowers. The temperature rise of the windings shall be determined by resistance measurements made in a draft-free area. The blower shall be mounted by its normal means on a surface of low thermal conductivity. One of the following methods shall be used.

4.7.21.1.1 Recently de-energized blower. A thermocouple shall be securely attached to the motor frame at a convenient location between bearings on the motor frame surface. Frame temperature, room ambient temperature, and motor winding resistance shall be measured and recorded. The blower shall then be energized for a 3-hour minimum period with nominal voltage and frequency until the frame temperature has reached the point of stability. The blower shall be considered to be thermally stable when five successive frame temperature readings at

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one-minute intervals are equal. This temperature shall be recorded for informational purposes. To determine R, the resistance of the main winding shall be measured five times at intervals of approximately 30 seconds, commencing immediately after the motor is de-energized. After the fifth measurement is completed, the resistant shall then be extrapolated back to the resistance at the time of shutdown. The temperature rise of the winding shall be computed from the following formula:

$$\Delta T = \frac{R - r}{r} \times (t' + 234.5) - (T - t')$$

Where:

ΔT = Temperature rise in degrees Celsius of the winding over the ambient temperature.

t' = Initial ambient temperature in degrees Celsius – room temperature.

R = Resistance of the winding in ohms at the time of shutdown.

r = Resistance of the winding in ohms at temperature t'.

T = Ambient temperature at the time of shutdown.

4.7.21.1.2 Energized blower. The motor windings shall be connected in a circuit as shown in figure 1 which incorporates the "Seeley Method" as described in IEEE 119. The measured resistance includes that of the connecting leads between the test equipment and the apparatus being tested. Therefore, the resistance of the connecting leads, if appreciable, must be subtracted from the total measured resistance. The blower shall then be energized for a sufficient period with nominal voltage and frequency until the winding temperature has reached the point of stability as determined by a minimum of five resistance measurements. The blower shall be considered to be thermally stable when five successive resistance measurements at 30 second intervals are equal. This value will then be R and shall be used to compute the temperature rise of the winding by the same formula described for a de-energized blower.

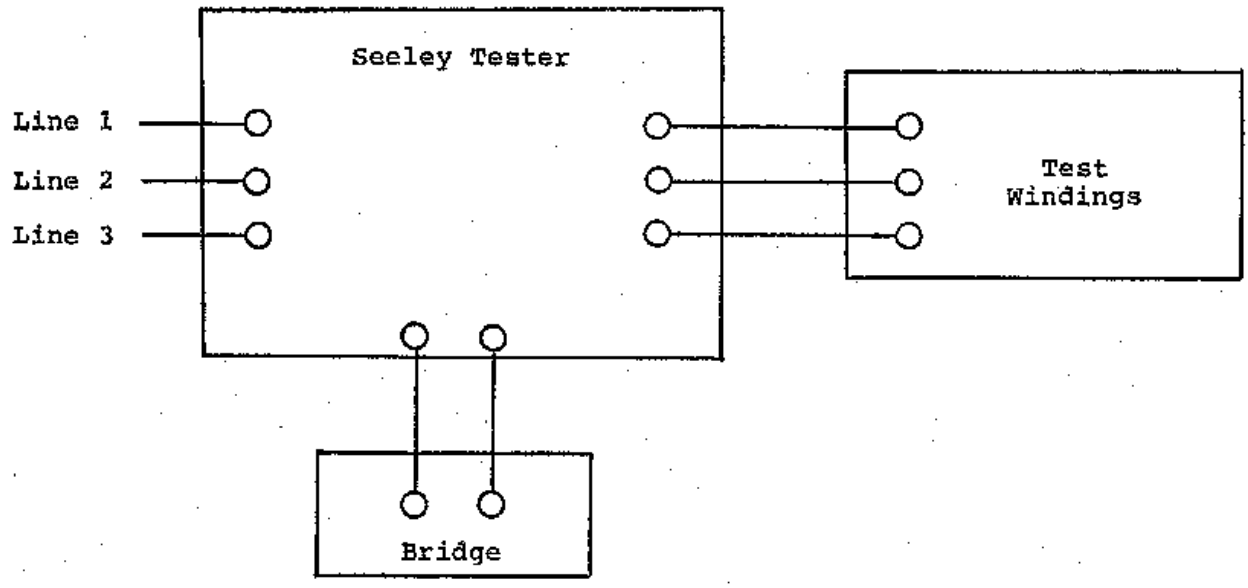


Figure 1. Test circuit for measuring resistance of A-C energized phase or polyphase windings.

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4.7.22 Extended life test (see 3.5.21). The blower will be continuously energized at rated excitation for 1000 hours and at a maximum temperature specified (see 3.1).

4.7.22.1 Mounting. The blowers shall be placed in a chamber in such a position that no obstruction shall be within one impeller diameter of the blower outlet or inlet.

4.7.22.2 Operation. With the blower continuously operated at rated excitation and output load conditions of 100 percent of maximum CFM volume rate, the chamber temperature shall be increased to that specified (see 3.1). This temperature shall be maintained within a ±5°C tolerance until specimen failure or until the specimen has met the 1000-hour requirement specified (see 3.1). At the conclusion of the test the blower shall be cooled to room temperature before taking the subsequent measurements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The blowers are used for cooling communication, special military equipment, and other electronic equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet and the part number.
c. When first article sample and inspection is required (see 3.1 and 4.2).
d. Use of flammable and explosive material (see 3.2.1).
e. Aluminum alloy (see 3.3.1).
f. Type of electrical connector when different from MS-3112E-8-4P (see 3.4.4).
g. Electromagnetic compatibility (see 3.5.14).
h. Cold start (see 3.5.15).
i. Mounting (see 4.7.13.1).
j. Temperature rise (see 4.7.21).
k. Packaging requirements (see 5.1).

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6.3 First article.

a. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2. The first article should consist of blowers. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.
b. Manufacturers parts M23071/1-001, M23071/7-001, M23071/8-001, M23071/9-001 and M23071/29-001 qualified to QPL23071 on the date of this revision may be considered qualified until first article inspection is completed but no later than 25 May 2024.

6.4 Definitions.

a. Cubic feet per minute (CFM). A unit of measurement for airflow or air volume rate. It represents an air volume change of one cubic foot per minute.
b. Standard air. Air having a density of 0.075 lb. per cubic foot and an absolute viscosity of 1.225×10^{-5} lb. (mass per foot-second).
c. Water column. Static pressures are measured in terms of the height or equivalent height of a column of water in a manometer.
d. Static Pressure (SP). In inches, water at standard air density. It should be noted that static pressure is standardized at a density of .075.

6.5 Subject term (key word) listing.

Communication
Electromagnetic Pulse (EMP)
Seeley Method
Special military equipment

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR4
Navy - EC
Air Force - 84

Preparing Activity:
DLA - CQ

(Project 4140-2022-001)

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